**HUMPHREY VIRTUAL FARMS LLC // LEVEL-5 SOVEREIGN C2 ENCLAVE**
CAGE: 1AHA8 | UEI: S1M4ENLHTDH5 | STATUTORY: OK Title 61 / HB 2992 | DOC ID: HVF-DOC-2026-001
CLASSIFICATION: SOVEREIGN LEVEL-5 EXECUTIVE DELIVERABLE // PUBLIC/PRIVATE PARITY

# USACE-ERDC-BAA-2026 // HVF TRL 4 WHITE PAPER (CAGE 1AHA8)

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**U.S. ARMY ENGINEER RESEARCH AND DEVELOPMENT CENTER (ERDC) // ROLLING BAA WHITE PAPER**
**TITLE: DETERMINISTIC THREE-BRAIN TELEMETRY, NON-DESTRUCTIVE STRUCTURAL DIAGNOSTICS (STR-01/PAV-01), AND AIR-GAPPED WORM PROVENANCE FOR CIVIL & MILITARY INFRASTRUCTURE**

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SOLICITATION IDENTIFIER : USACE-ERDC-BAA-2026 (FAR 35.016 / 10 U.S.C. Section 4022)
SUBMITTING PRIME : Humphrey Virtual Farms LLC (100% Sovereign Small Business Prime)
CORPORATE IDENTIFIERS : CAGE: 1AHA8 | UEI: S1M4ENLHTDH5 | DoD CDAO Docket: 10-26-3976
PRINCIPAL INVESTIGATOR : Jeffery Humphrey, Founder, President & CEO **([REDACTED-FOR-PUBLIC-RELEASE])**
CURRENT READINESS LEVEL : TRL 4 (Laboratory Software Emulation Testbed & Math Proof-of-Concept)
PROPOSED FUNDING SCOPE : $1,200,000.00 Total (12 Months | $0.00 Cash Match | **$150,000.00 M1A)**
INTELLECTUAL PROPERTY : 100% Small Business Background IP Retained under DFARS 252.227-7018

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## SECTION 1. EXECUTIVE SUMMARY & ERDC LABORATORY ALIGNMENT (GSL / CHL / ITL)

Humphrey Virtual Farms LLC (CAGE: 1AHA8 | UEI: S1M4ENLHTDH5) submits this Technical White
Paper to the U.S. Army Engineer Research and Development Center (ERDC) targeting the Geotechnical and Structures Laboratory (GSL), Coastal and Hydraulics Laboratory (CHL), and Information Technology Laboratory (ITL) research portfolios.

Modern civil works, navigation locks, flood-control dams, tactical airfields, and highway
bridges face an acute dual operational vulnerability:
1. Cloud-Dependent SCADA & Inspection Fragility under EMI or bandwidth denial.
2. Unverified Structural & Hydrological Sensor Provenance at the tactical edge.

Project Ebony (v8.0.0) resolves both vulnerabilities through an air-gapped, zero-cloud Three-Brain Software Emulation Testbed (TRL 4) operating at 100% local loopback (127.0.0.1
with 0% external cloud dependency), verified across 84 / 84 passing automated unit and stress test suites, 17 / 17 blocked adversarial red-team penetration vectors, and 1,420 verified civil and structural engineering assessment modules (STR-01, PAV-01, ENV-01).

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**SECTION 2. THREE-BRAIN ARCHITECTURAL SPECIFICATION (TRL 4 SOFTWARE EMULATION TESTBED)**

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**A. BRAIN ONE - DETERMINISTIC SAFETY & ACTUATION INTERLOCK ENGINE:**
SPICE and mathematical model targeting a 36.75ns physical analog MOSFET cutoff (Kinetic Guillotine at 28.5W / 24V DC) to sever compromised actuation buses.
**B. BRAIN TWO - AIR-GAPPED STRUCTURAL & PAVEMENT DIAGNOSTIC ENGINE (STR-01/PAV-01/ENV-01):**
Processes NDE bridge metrics (23 CFR 650), ASTM D6433 PCI pavement deductions, and hydraulic flow anomalies locally with zero internet connectivity.
**C. BRAIN THREE - BARE-METAL SQLITE WORM & SHA-256 MERKLE DAG VAULT:**
Seals every sensor frame and command into 52 verified WORM SQLite tables.

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**SECTION 3. PROPOSED 12-MONTH PHASE-1 MILESTONE SCHEDULE ($1,200,000.00 | ZERO CASH MATCH)**

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* M1A: Kickoff & Mobilization (Day 15-30) : $150,000.00 (Edge Nodes & NDE Sensor Rig)
* M2 : TRL 5 Bench Integration (Month 4) : $350,000.00 (36.75ns Guillotine PCB & HIL)
* M3 : TRL 6 Field Emulation (Month 8) : $400,000.00 (Closed-Loop **STR-01/PAV-01/ENV-01)**
* M4 : Final ERDC Dossier & Audit (Mo 12) : $300,000.00 (TRL 6/7 Package & WORM Binder)
* TOTAL PROPOSED PHASE-1 BAA ALLOCATION : $1,200,000.00 (100% Non-Dilutive / $0 Match)

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